

Person Name and Information

Education

Georgia Institute of Technology, Atlanta, GA
Bachelor of Science in Computer Science

May 2028
GPA: 3.75/4.00

Relevant Coursework: Linear Algebra, Introduction to Object-Oriented Programming, Data Structure and Algorithms, Computer Organization and Programming

Technical Skills

Computing Languages - Python, Java, JavaScript, C#

Machine Learning - PyTorch, TensorFlow

Web Development - React JS, Express.js, Flask, RESTful API, HTML, CSS, Tailwind CSS

Data Science & Scientific Computing - NumPy, Pandas, Numba, High-Performance Computing

Work Experience

University Summer Research Program, [State/Region]

July 2024 - August 2024

Student Researcher

- Conducted research under a university professor
- Contributed to improving precipitation estimates around synthetic tropical cyclone tracks by training generative AI models on historic U.S. East Coast cyclone data
- Presented research at national earth science conference

Programming Projects

YouTube Creator Assistant Project

Core Developer

- Engineered an AI-powered platform to support content creation workflows integrating OpenAI GPT-4 for video script generation and DALL-E 3 API for video thumbnail creation, with Sharp library for image processing and compositing
- Built full-stack application with Express.js RESTful API (file uploads, async processing) and React/Tailwind CSS frontend

Home Disease Detector Project

Head Developer

- Developed a deep learning-based website that diagnoses diseases by analyzing user-reported symptoms through a series of questions
- Won 2nd place in hackathon with 162 participants for this project

Leadership and Community Engagement

AI Bot Programming Competition Club, Lead Engine Developer

December 2025 - Present

- Created reference implementation for competitors on how to accelerate the game engine to enable faster move simulations for their AI bots
- Achieved 26x performance speedup on game board simulation by integrating Numba JIT compilation and converting class-based Python board code to jitclass-optimized data structures

School AI/ML Club, ML Analysis Team Member

September 2025 - Present

- Designed the algorithms to collect Southeast Asia flood training data: flood and non-flood event locations and the historic precipitation and soil moisture around those event areas
- Used PyTorch to build and train CNN-LSTM models on the PACE computing clusters
- Achieved 77.1% accuracy and 74.3% recall for 5-day advance flood prediction

Student Hackathon Organization, Programming Committee Co-director

March 2023 - February 2025

- Led Programming Committee members in planning hackathon workshops, activities, event schedule, and prizes
- Oversaw the 2025 hackathon event at a local university as one of the directors

Additional Skills and Interests

Sports: Basketball, Volleyball

Interests: Mathematics, Artificial Intelligence, Cybersecurity, Sketching, Painting